DECENT COACHING CLASSES

X Semi

Mind Your Math - IX

ANSWER KEY & MARKING SCHEME

Time: 1 Hour 30 Minutes **Maximum Marks: 30**

Q.1

Q.1 (A) [1 mark each]

(1) **(a) 5** – $(2 \times 4) - (3 \times 1) = 8 - 3 = 5$

(2) **(b) $x^2 + 2x + 1 = 0$** – the only equation of degree 2

(3) **(b) 3** – $d = 5 - 2 = 3$

(4) **(a) AC/PR** – corresponding sides of similar triangles are in the same ratio

Q.1 (B) [1 mark each]

(1) **12** ($d = 3$)

(2) $x - 2 = 0$ or $x - 3 = 0 \Rightarrow$ **$x = 2, x = 3$**

(3) $D = (2 \times 1) - (3 \times 1) =$ **-1**

(4) Ratio of areas $= (1/2)^2 =$ **1 : 4**

Q.2

Q.2 (A) [2 marks each – any two]

(1) $D = (1)(-1) - (1)(1) =$ **-2** [½]; $D_x = (5)(-1) - (1)(1) = -5 - 1 =$ **-6** [½]; $D_y = (1)(1) - (5)(1) = 1 - 5 =$ **-4** [½]; $x = -6/-2 =$ **3**, $y = -4/-2 =$ **2** [½]

(2) $a =$ **2**, $d =$ **2** [½]; $t_n = 2 + (10 - 1) \times 2 = 2 + 18 =$ **20** [1½]

(3) $x^2 -$ **3**$x -$ **2**$x + 6 = 0$ [½]; $x(x -$ **3**$) - 2(x -$ **3**$) = 0$ [½]; $(x -$ **3**$)(x -$ **2**$) = 0$ [½]; **$x = 3$ or $x = 2$** [½]

Q.2 (B) [2 marks each – any four]

(1) Adding: $2x = 10 \Rightarrow$ **$x = 5$** [1]; $5 + y = 7 \Rightarrow$ **$y = 2$** [1]

(2) $x^2 - 7x + 12 = (x - 3)(x - 4) = 0$ [1] $\Rightarrow$ **$x = 3$ or $x = 4$** [1]

(3) $a = 1, d = 1, n = 10$; $S_n = (n/2)[2a + (n - 1)d] = 5[2 + 9]$ [1] = **55** [1]

(4) DE ∥ BC $\Rightarrow$ AD/DB = AE/EC (BPT) [1]; $2/3 = 4/EC \Rightarrow$ **EC = 6** [1]

(5) (i) $(5 \times 4) - (3 \times 2) = 20 - 6 =$ **14** [1]; (ii) $(-1 \times 4) - (2 \times 3) = -4 - 6 =$ **-10** [1]

Q.3 [3 marks each – any two]

Q.3

(1) $D = (2)(1) - (1)(1) =$ **1** [1]; $D_x = (5)(1) - (1)(3) =$ **2**, $D_y = (2)(3) - (5)(1) =$ **1** [1]; $x = D_x/D =$ **2**, $y = D_y/D =$ **1** [1]

(2) $x^2 + 4x + 2x + 8 = 0$ [1]; $x(x + 4) + 2(x + 4) = 0 \Rightarrow (x + 4)(x + 2) = 0$ [1]; **$x = -4$ or $x = -2$** [1]

(3) $t_8 = a + 7d = 5 + 7 \times 3 =$ **26** [1½]; $S_8 = (8/2)(5 + 26) = 4 \times 31 =$ **124** [1½]

Q.4 [4 marks – any one]

Q.4

(1) Ratio of areas of similar triangles = (ratio of corresponding sides)² [1]; $16/25 = (8/x)^2$ [1] $\Rightarrow 4/5 = 8/x$ [1] $\Rightarrow$ **$x = 10$ cm** [1]

(2) Let the numbers be x and y. $x + y = 12$ … (I), $x - y = 4$ … (II) [1]; adding (I) and (II): $2x = 16$ [1] $\Rightarrow$ **$x = 8$** [1]; $y = 12 - 8 =$ **4**. The numbers are **8 and 4**. [1]

— End of Answer Key —